Decay-Seamed Block Thomas v0 - review 1

As a **quick sanity-check script**, it's useful. As a **serious tester/validator of a Decay-Seamed Block Thomas (DSBT) algorithm**, it's quite limited.

What it does well

The script checks three ideas:

1. **Theoretical decay constant**
   - Computes $\rho = 2 - \sqrt{3} \approx 0.267949$.
   - Prints ρ^s for several seam widths.
   - This is relevant because DSBT relies on exponential decay across block boundaries.
2. **Empirical inverse decay**
   - Builds a natural-spline tridiagonal system on a nonuniform knot sequence.
   - Inverts the matrix and measures decay of a central column of A^{-1} .
   - Fits a log-slope to estimate an empirical decay factor.
3. **Back-of-the-envelope parallelism model**
   - Estimates number of blocks, seam overhead, and thread work.
   - Gives a rough speedup intuition.

Those are all reasonable checks for the *motivation* behind a decay-seamed method.

What it does not test

The major issue is that it never actually runs a Decay-Seamed Block Thomas solver.

The script:

- builds a matrix,
- computes a full inverse,
- measures decay,
- estimates workload,

but nowhere does it:

- partition the system into blocks,
- apply seam halos,
- solve local block systems,
- stitch blocks,
- compare DSBT output against exact Thomas output.

So it validates an assumption of DSBT (inverse decay), not the algorithm itself.

I would call it:

> **a DSBT feasibility demonstration, not a DSBT correctness test.**

Missing correctness tests

A real DSBT tester should verify:

1. Solution accuracy

Generate many tridiagonal systems:

$$Ax = b$$

and compare:

- Exact Thomas solution x_T
- DSBT solution x_D

using

$$\|x_D - x_T\|_\infty$$

and

$$\frac{\|x_D - x_T\|_2}{\|x_T\|_2}$$

for varying seam widths s .

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2. Error-vs-seam-width curve

The core DSBT prediction is

$$\text{error} \sim \rho^s.$$

You should test:

```
s = 5
s = 10
s = 15
s = 20
s = 25
s = 30
```

and verify that the measured error decreases exponentially.

That is arguably the single most important validation.

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3. Boundary-seam error localization

DSBT predicts that most error is concentrated near seams.

Plot

$$|x_D - x_T|$$

across the domain.

You should see spikes near block interfaces and negligible error elsewhere.

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4. Nonuniform-grid stress tests

The current script only uses one coefficient variation level:

```
CV ~ 0.39
```

from the random spacing generation.

A stronger tester would vary:

```
CV = 0.0
CV = 0.1
CV = 0.3
CV = 0.5
CV = 1.0
```

and verify that decay and accuracy remain acceptable.

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5. Pathological matrices

Test:

- nearly singular systems,
- highly varying diagonal dominance,
- alternating coefficients,
- clustered knots,
- random coefficient distributions.

These often reveal seam-instability issues.

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6. Runtime measurement

The script estimates work analytically.

A proper tester should measure:

```
time_thomas
time_dsbt
speedup = time_thomas / time_dsbt
```

for increasing problem sizes.

Without timings, performance claims remain theoretical.

One methodological concern

This section:

```
G = np.linalg.inv(A)
```

computes the entire inverse.

For $n = 4096$ that's fine, but for validating DSBT it is somewhat indirect.

A stronger decay test would estimate selected inverse-column entries through solves rather than constructing the full dense inverse. That scales better and resembles how DSBT would actually be used.

Overall assessment

If the goal is:

- ”Does the decay assumption behind DSBT appear valid?”

then the script is **good (about 7/10)**.

If the goal is:

- ”Does my Decay-Seamed Block Thomas implementation work correctly and achieve the predicted error/runtime behavior?”

then the script is **poor (about 3/10)** because it never exercises the algorithm itself.

The biggest missing component is a direct **DSBT-vs-Thomas accuracy benchmark across seam widths and block sizes**. That would turn this from a conceptual validation script into a genuine algorithm test harness.